

2.2.9 Combo Drive

This is a drive that can read and write CDs and also read DVDs.

2.2.10 Construction

All optical media are prepared using the same procedure. The substrate is the clear plastic layer that forms the bulk of the disc. A thin layer of dye is placed above the plastic. This is where the laser records data in the form of burned or clear spots. The dye, which is commonly made of Cyanine or a compound thereof, changes its optical properties under the laser beam. In the case of pre-manufactured CDs, the pattern of pits and lands are stamped on the substrate itself. A reflective film, usually made of silver or gold, comes next. Finally, a protective acrylic coat is applied. Additionally, a layer for labelling purposes is added. In case of dual-layer media, there are two layers of dye with a layer of semi-reflective material in between them.

2.2.11 DAO

The Disk-at-Once mode of writing refers to when all tracks are recorded without stopping the laser; the disc is finalised at the end and no further data can be added. In a DAO writing, there is only one Lead-in and Lead-out.

2.2.12 DVD

The Digital Versatile Disc was created to fulfil the need for storage of high-quality video data. Two camps existed in the DVD technology arena: Philips and Sony were developing a new technology for storage of high-quality video, while simultaneously, Toshiba, Hitachi, and others were involved in developing a parallel technology that was incompatible with Sony's technology. The two camps agreed to collaborate and the result was the DVD, in 1995.

DVDs can store up to six times as much as a CD, at 4.7 GB (if you consider a Kilobyte as 1,000 bytes), while using a disk of the same size. This was possible by using a laser of a smaller wavelength—630 nanometres—and reducing the width of the tracks and the distance between them. The original DVD drive had a data transfer rate of 1.5 Megabytes per second.